

CO1-ASSESSMENT TOOL-OUTPUT PREDICTION

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COURSE CODE & NAME:CSA0307-DATA STRUCTURES

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Q1 Missing Element

1 mark

Write a C program for Find Missing Element in an Array

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int n,i,sum=0,arraysum=0,missing;
5     printf("Enter the value of n:");
6     scanf("%d",&n);
7     int a[n-1];
8     printf("Enter %d elements:\n",n-1);
9     for(i=0;i<n-1;i++)
10 {
11     scanf("%d",&a[i]);
12     arraysum=arraysum+a[i];
13 }
14 sum=n*(n+1)/2;
15 missing=sum-arraysum;
16 printf("Missing element=%d",missing);
17 return 0;
18 }
19
```

Input

```
5
1 2 4 5
```

Output

```
Enter the value of n:Enter 4
elements:
Missing element=3
```

Run

Save

Submit

Q3 Largest and Smallest element

1 mark

Write a C program for Find the Largest and Smallest Element in an Array

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[5]={10,5,25,2,13};
5     int i,largest,smallest;
6     largest=smallest=a[0];
7     for(i=1;i<5;i++)
8     {
9         if (a[i]>largest)
10            largest=a[i];
11         if(a[i]<smallest)
12            smallest=a[i];
13     }
14     printf("Largest=%d\n",largest);
15     printf("Smallest=%d\n",smallest);
16     return 0;
17 }
18
```

 Run Save Submit

Input

10 5 25 2 13

Output

System error.

Write a C program for Searching an Element in a Sorted Array

Your Code

```
1 #include<stdio.h>
2 int main()
3 {
4     int a[10],n,key,low,mid,i;
5     printf("enter the number of elements:");
6     scanf("%d",&n);
7     printf("Enter elements in sorted order");
8     for(i=0;i<n;i++){
9         scanf("%d",&a[i]);
10    }
11    printf("Enter element to search:");
12    scanf("%d",&key);
13    low=0;
14    high=n-1;
15    while(low<=high){
16        mid=(low+high)/2;
17        if(a[mid]==key){
18            printf("Element found at position %d",mid+1);
19            return 0;
20        }
21        else if (a[mid]<key){
22            low=mid+1;
23        }
24        else{
25            high=mid-1;
26        }
27    }
28    printf("Element not found");
29    return 0;
30 }
```

Input

stdin input

Output